

Q 3 (i) (m)

A 3 (i) (or)

In biology, domain means largest of all groups in the classification of life. Domain is the group of all kingdoms or ~~above all~~ taxonomic category above all kingdoms. In 1990, Carl Woese introduced a three domain system of classification. The three domain system are :-

Archaea.

Bacteria

Eukarya.

Classification into 3 domain is based on the difference of the sequence of nucleotides in the rRNA (ribosomal ribonucleic acid) of the cell, cell membrane lipid structure and its sensitivity to antibiotics.

Domain Archaea :-

Following are the characteristics of domain Archaea :-

Domain Archaea are prokaryotic cell. Their cell wall contains no peptidoglycan. Their rRNA (ribosomal RNA) is not found in bacteria and Eukarya.

④ Archea are not sensitive to some antibiotics that affect the bacteria, they are sensitive to some antibiotics that affect the eukarya.

⑤ Archea often live in extreme environment

⑥ Archea membrane can withstand higher temperature and stronger acid concentration

⑦ Archeal creatures that include:-

① Methanogens

② Halophiles

③ Thermacidophiles.

* Domain Bacteria:-

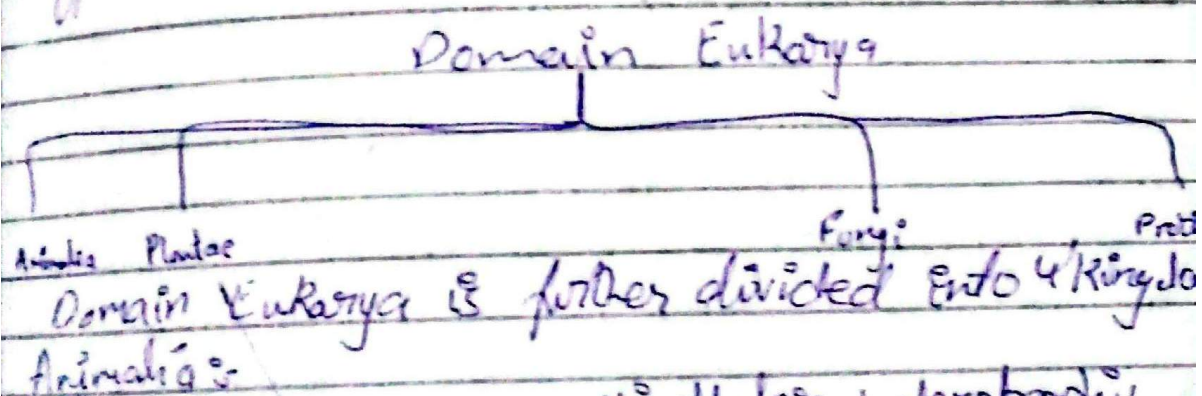
Following are the characteristics of domain Bacteria are:-

1. Domain Bacteria are prokaryotic cells.
2. Their cell wall contains peptidoglycan.
3. ~~Their RNA~~ They have rRNA that is unique to Bacteria.
4. Domain Bacteria are sensitive to some traditional antibacterial antibiotics but are resistant to some antibiotics that affect the eukarya.

Domain Eukarya:-

Following are the characteristics of domain Eukarya are:-

- ① Domain Eukarya have eukaryotic cells.
- ② Not all eukarya have cell wall, their cell wall does not contain peptidoglycan.
- ③ Their rRNA is unique to eukarya.
- ④ Eukarya are resistant to some transitional antibacterial antibiotics, these are sensitive to some antibiotics that affect the eukaryotic cells.



Animals:-

It is a multicellular heterotrophic organisms. which can generally move from place to place. They can hunt their own food. They are called producers. There are vertebrates :- insects, starfish etc. and invertebrates :- humans, frogs etc.

Plantae :-

It is multicellular autotrophic organisms which will create their own food. for example mango tree etc.

Fungi :-

It is multicellular mostly organisms which are heterotrophic. like mushroom.

Protista :-

It is unicellular eukaryotic autotrophic and heterotrophic organism.

Q 3(ii)
A: 3(ii)

The nucleus is the control centre of the cell, containing the genetic information (DNA) and directly controls cellular activities.

① Nuclear envelope:-

Double membrane surrounding the nucleus. outer membrane is continuous with rough endoplasmic reticulum. controls entry / exit of materials.

② Nuclear pores:-

Small, openings in the nuclear envelope that allow rRNA, ribosomes and other molecules to move b/w nucleus and cytoplasm.

③ Nucleoplasm:-

fluid matrix inside the nucleus in which chromatin and nucleolus is suspended.

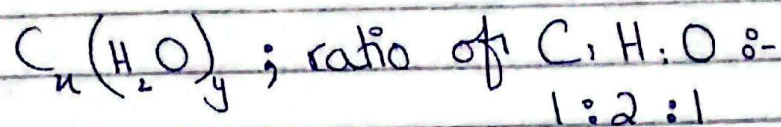
④ Nucleolus:-

Dense body inside nucleus site of rRNA synthesis which combines with protein to form ribosomes.

Q2 33653251

Q. 3 (iii) (or) _____
A. 3 (iii) (or) _____

Carbohydrates are classified on the basis of the numbers of saccharide (monosaccharide) units in the molecule. General formula :-



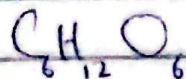
It's classified into three types :-

* Monosaccharide :-

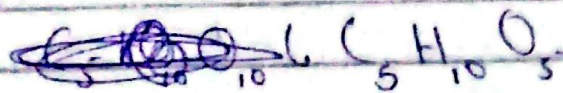
It is the simplest carbohydrates, it is a single saccharide unit; cannot be hydrolysed further, Sweet and crystalline and it is soluble in water. Example:- Glucose, fructose, galactose, ribose etc.

Molecular formula :-

Glucose :-



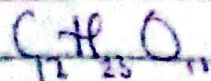
Ribose :-



* Disaccharides :-

It is form by two monosaccharides bonded by condensation (releasing water). It is slightly less sweet.

Molecular formula :-



Sucrose :- cane sugar

Glucose + Fructose



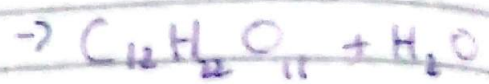
Maltose :- Malt sugar

Glucose + Glucose



Lactose :- (milk sugar)

Galactose + glucose



Q: 3(iv) (05)

A: 3(iv) (05)

Effect of temperature :-

As temperature increases from low to optimum, enzyme activity increases because molecules move faster, optimum temperature ~~increasing collision frequency b/w enzyme~~

:- $36^{\circ}C$ - $38^{\circ}C$ (body temperature)

Enzyme works fastest here

Above optimum :-

Atoms of enzymes vibrate vigorously, globular structure and active site are damaged. This is called denaturation. Activity falls sharply to 0.

At very low temperature :-

Enzymes is inactive due to insufficient activation energy.

Activity resumes when normal temperature is restored.

Effect of substrate concentration:-
When enzymes molecule are in ~~excess~~ excess, increasing substrate concentration increases reaction rate because more substrate molecules fill ~~is~~ the available active site.

* Up to a certain point, rate is directly proportional to substrate concentration.

* Beyond a maximum :- All active sites are occupied called saturation of active sites. Further addition of substrate has no effect on rate.

Section - B

Q: 2 (i) (or)
A: 2 (i) (or)

Medicine / surgery :-

- It's degree is MBBS, it diagnoses and treats human disease and disorders

Biotechnology:- Studies organisms to develop useful products (vaccines, medicines etc)

Agriculture / farming :-

Applies biological knowledge to improve crop yield & fertility for food.

Q: 2 (ii) (or)
A: 2 (ii) (or)

Ronald Ross :- In 1897, A British physician who fed female Anopheles mosquitoes on malaria-infected patients, then dissected the mosquitoes and found Plasmodium in their stomach walls. His hypothesis is the Anopheles mosquito is the vector transmitting the parasite.

Q: 2(iii)(or)

A: 2(iii)(or)

Domain Archaea :-

Following are the characteristic of domain Archaea :-

- i) Domain Archaea are prokaryotic cells.
- ii) Their cell wall doesn't contain peptidoglycan.
- iii) Their rRNA does not found in Bacteria and Eukarya.
- iv) They often live in extreme environment.

In 1990, Carl Woese introduced a domain system which ~~introduced~~ include:

- 1) Archaea 2) Bacteria 3) Eukarya.

Q: 2 (iv)(or)

A: 2 (iv)(or)

~~Chloroplast~~ Chloroplast

It has double membrane & inner membrane form thylakoids groups into grana & fluid filled stroma outside.

It's function is photosynthesis & convert light energy into chemical energy.

Mitochondria

It has double membrane & inner membrane form cristae & fluid filled matrix inside.

ATP production & release energy from glucose.

Q1- is found in plant cell only

It is found in eukaryotic cells.

Q2(v)

A: 2(v)

Metaphase :-

Chromosomes move to the equatorial plate (middle of cell). Spindle fibres from both Poles attach to the centomere of each chromosome, aligning them precisely. This is the stage where chromosomes are more visible.

Anaphase :-

The centomere split & separating each chromosome into two Chromatids. Spindle fibres shorten and pull the chromatids to opposite poles of the cell. By the end of the anaphase each pole has a complete set of chromosomes.

Q: 2(vi)

A: 2(vi)

Prophase I is the longest phase of meiosis.

Chromatin condenses into chromosomes and the nuclear membrane begins to break down.

Synapsis :- homologous chromosomes come together and pair up along their entire length.

Each pair of homologous chromosomes form a bivalent (tetrad of 4 chromatids) :- 2 chromosomes $\times$ 2 chromatids each.

Crossing over :- non-sister chromatids of homologous chromosomes exchange segments at points called chiasmata, creating new gene combinations and contributing to genetic variation.

Q: 2(vii)

A: 2(vii)

Homeostasis :-

is the tendency of living organisms to maintain a stable, relatively constant internal environment.

Q: 2 (viii)
A: 2 (viii)

Shoot system :

Located above the ground.
Consists of stem, leaves and reproductive organs (flowers, fruits, seeds). Functions include photosynthesis (leaves), transport of food and water through xylem and phloem and reproduction.

Root system :

Located below ground.
Consist of primary and secondary roots.

Q: 2 (in) (or)
A: 2 (in) (or)

① Energy storage :-

Lipid store. Long term energy in fat cell and liver.

② Cell membrane component :-
Phospholipids and cholesterol form the plasma membrane bilayer.

③ Hormone synthesis :-

Steroid lipids form hormones ~~and~~ like testosterone and aldosterone.

Insulation :-

- lipid acts as insulators against atmospheric heat and cold.

Q: 2(n)(00)

A: 2(n)(or)

A nucleotide is the monomer of nucleic acids (DNA and RNA).
It has 3 components :-

- ① Pentose sugar
- ② Nitrogenous base.
- ③ phosphate group.

Q: 2(ni)(00)

A: 2(ni)(or)

Many enzyme require a non-protein helper a cofactor. Three types :-

① Activator :-

Detachable inorganic ions that activate the enzymes

② Prosthetic group :-

An organic molecule ~~permanently~~ ~~permanently~~ Permanently and tightly attached to the enzyme.

Coenzyme :-

A detachable organic cofactor that temporarily binds to the enzyme.

MCQs:-

(1) C

(2) C

(3) B

(4) C

(5) B

(6) C

(7) B

(8) B

(9) C

(10) C

(11) B

(12) C